Teacher Wu Kuan | Exam at a Glance #16 — Vector Addition: Head to Tail

English lecture notes · YouTube Short 20acn63eQ3U

1. The trap question

(3, 1) + (1, 2) = ? And how long is it?

2. The method

• Graphically: put the tail of the second at the head of the first.
• With coordinates: add x's and y's → (4, 3).
• Length by Pythagoras: $\sqrt{(4^2 + 3^2)} = 5$.

3. Parent check method

Ask your child to walk it on grid paper: right/up for the first, then right/up for the second.

Practice

1. a = (2, 1), b = (1, 3). Find a + b.
2. Continuing, what is the length of a + b?
3. a = (5, 2), b = (−1, 1). Find a + b and its length.
4. a = (4, −1) and a + b = (6, 2). Find b.
5. a = (1, 2). Find 2a + (3, −1).

Answers

1. (3, 4)
2. 5
3. (4, 3), length 5
4. (2, 3)
5. (5, 3)

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